Facial Micro-Feature Detection Model Documentation

1. How the Model is Built

The model is built upon a sophisticated pipeline that transforms raw video data into high-dimensional feature sequences, which are then processed by a hybrid deep learning architecture.

A. Feature Extraction Pipeline

Instead of using raw pixels, we extract interpretable, high-level features from facial landmarks to focus on "micro" movements.

- **MediaPipe FaceMesh**: We detect 478 standardized 3D facial landmarks for every video frame.
- **166-Dimensional Feature Vector**: For each frame, we compute 166 specific features:
    - **Surface Vectors (Novelty)**: We perform Delaunay Triangulation on key facial regions (Brow, Cheek, Eye, Jaw, Lips, Mouth). We calculate the **Magnitude** (stretch), **Variance** (deformation texture), and **Directional Angle** of these triangles to capture skin movements that simple points miss.
    - **Geometric Features**: Standard distances like Eye Aspect Ratio (EAR) and Mouth Opening.
    - **Kinematic Features**: Velocity and Acceleration of movements using temporal buffers.
    - **Frequency Domain**: Fast Fourier Transform (FFT) features to detect high-frequency tremors relevant to Parkinson's.

B. Data Preprocessing

- **Normalization**: Landmarks are normalized relative to the nose tip and face width to be invariant to camera distance and head rotation.
- **Scaling**: Input features are standardized (Z-score normalization) to ensure the model treats all features equally (e.g., small eye movements vs. large jaw movements).
- **Sequence Padding**: Since videos have different lengths, we use padding to create uniform sequence batches for the model.

C. Model Architecture: AdvancedCNNLSTM

The model combines three powerful neural network paradigms:

1. **Residual 1D CNN Encoder**:
    - Consists of **Residual Convolutional Blocks**.
    - Extracts *short-term* local temporal patterns (e.g., a sudden twitch typically lasting few frames).
    - Uses **Residual Connections** to allow easier training of deeper layers without losing signal information.

2. **Bi-Directional LSTM (Long Short-Term Memory)**:
   - Processes the sequence in both forward and backward directions.
   - Captures *long-term* dependencies and context (e.g., how an expression evolves from onset to peak to offset).
3. **Multi-Head Self-Attention Mechanism**:
   - Unlike standard RNNs that treat all time steps sequentially, Attention allows the model to "focus" on the most relevant frames in the video, ignoring neutral or noisy frames.
   - **Multi-Head** means it can focus on different aspects simultaneously (e.g., Head 1 watches the eyes, Head 2 watches the mouth).

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2. Why We Use These Techniques

Technique	Rationale & Advantage
Surface Vectors	Traditional landmarks (dots) only track position. Micro-expressions involve **skin deformations** (wrinkling, bulging) between the dots. Surface vectors quantify this skin texture change, which is critical for detecting subtle Parkinsonian symptoms.
Hybrid CNN-LSTM	**CNNs** are excellent at detecting "what" happened (local patterns), while **LSTMs** are excellent at understanding "when" and "how" it evolved over time. Combining them gives the best of both worlds for temporal pattern recognition.
Residual Connections	Deep networks often suffer from "vanishing gradients" where they stop learning. Residual connections provide a "highway" for information to flow, ensuring stable and effective training.
Self-Attention	A video might be 5 seconds long, but the micro-expression only lasts 0.2 seconds. Attention mechanisms allow the model to automatically weight the important 0.2s heavily while disregarding the irrelevant 4.8s, significantly improving accuracy.
Frequency (FFT)	Parkinson's disease is characterized by tremors. FFT features specifically look for these rhythmic, high-frequency oscillations that geometric distance features might miss.

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3. How the Model Processes and Predicts

The prediction process (Inference Pipeline) follows these steps:

1. **Input**: The user uploads a video file (e.g., .mp4).
2. **Frame Extraction**: The video is decoded into individual frames.
3. **Landmark Detection**: MediaPipe finds the 478 mesh points on the face.
4. **Feature Computation**:

- The system calculates the 166 features (Surface Stats, Velocity, etc.) for the current frame.
- It compares current landmarks with previous ones to determine motion.
- This results in a sequence of shape

(T, 166), where T is the number of frames.

5. **Normalization**: The scaler loaded from

scaler.pkl transforms the data to match the training distribution.

6. **Model Inference**:
   - **CNN** reduces the feature dimensionality and extracts local patterns.
   - **LSTM** analyzes the temporal flow.
   - **Attention** aggregates the most critical information into a single "Context Vector".
   - **Classifier** (fully connected layers) maps this vector to the 8 emotion classes.
7. **Output**: The model outputs a probability distribution (e.g., *85% Happy, 10% Neutral*). The highest probability is returned as the predicted class.